



**Verified Carbon
Standard**

NOOR 2 ABU DHABI STREET LIGHTING PROJECT



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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Noor 2 project represents a significant step forward in Abu Dhabi's commitment to sustainability and energy efficiency. As part of the Emirate's broader strategy to reduce greenhouse gas emissions and transition towards cleaner energy, the project focuses on modernizing the road lighting infrastructure with advanced, energy-efficient technology. The project aims to significantly reduce energy consumption and minimize the reliance on fossil fuels, directly supporting the UAE's goal of achieving Net Zero emissions by 2050.

The Noor 2 project involves a long-term concession awarded by Department of Municipalities & Transport (DMT) to Nojoom with the aim of enhancing Abu Dhabi's road lighting infrastructure. Under this agreement, Nojoom will be responsible for the supply, installation, financing, operation, and maintenance of 133,473 existing high-pressure sodium (HPS) and metal halide (MH) luminaires, replacing them with advanced, energy-efficient LED luminaires. Additionally, the LED lights will be wirelessly connected to the central road lighting control system, resulting in improvements to operational efficiency.

This project is in Abu Dhabi and will cover a vast network of both urban roads and highways across the mainland. The project is developed as a grouped project, divided into five distinct phases, with each phase targeting a different geographical zone within the Emirate. This structured approach ensures that the project is delivered in a way that maintains operational efficiency.

The project's LED lights are far more energy-efficient compared to traditional HPS and MH luminaires that exist prior to project implementation, leading to a substantial reduction in electricity consumption. This decrease in energy demand is expected to generate significant greenhouse gas emission reductions, as the reliance on fossil fuel-based electricity is minimized. Over the crediting period, the project is expected to achieve almost 2,650 million kWh of electricity savings. These savings will contribute to an average GHG emission reduction of approximately 102,543 tons of CO₂ annually.

The project activity instance is defined as a particular set of implemented technologies and/or measures that constitute the minimum unit of activity necessary to comply with the criteria and procedures applicable to the project activity under the methodology applied to the project in the VCS Program Definitions v4.5. Hence, each project activity instance refers to each phase of the project.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	14-February-2025	VCS	KBS Certification Services Limited	N/A

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	Energy Demand
Project activity type	Lighting Efficiency Improvement Project

1.4 Project Eligibility

1.4.1 General eligibility

This project falls under the category of energy-efficient improvements in lighting systems, specifically the replacement of electric lighting with more energy-efficient alternatives, such as the replacement of incandescent bulbs with compact fluorescent lights (CFLs) or light-emitting diodes (LEDs).

The project consists of 5 phases with each phase being a project activity instance, and each phase staying below required thresholds provided by VCS, thereby ensuring full compliance with VCS Program requirements. The methodology AMS-II.C. Demand-side energy efficiency activities for specific technologies Version 16.0 is approved under CDM Program, which is a VCS approved GHG program.

1.4.2 AFOLU project eligibility

Not applicable as this is not an AFOLU project.

1.4.3 Transfer project eligibility

Not applicable as this is not a transfer project or a CPA.

1.5 Project Design

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.5.1 Grouped project design

This project is designed as a grouped project, comprising 5 project activity instances, with each instance representing a distinct phase of the project. It covers the Abu Dhabi Mainland, including both urban roads and highway areas, and is subdivided into five phases, each consisting of smaller geographical zones to ensure the project is developed efficiently.

1.6 Project Proponent

Organization name	Nojoom Project Company Streetlighting Installation Repair and Maintenance LLC
Contact person	Lukasz Wiecek
Title	General Manager
Address	9th floor, East Tower, Abu Dhabi Mall, PO Box 7075, Abu Dhabi, United Arab Emirates
Telephone	+971 56 526 1205
Email	lukasz.wiecek@nojoom.me

1.7 Other Entities Involved in the Project

Organization name	elementsix
Role in the project	Consultant
Contact person	Amer Arafat
Title	Partner
Address	Technohub 1, DTEC, Dubai Silicon Oasis, Dubai, UAE
Telephone	+971 52 938 9590
Email	amer@element6.cc

1.8 Ownership

The ownership of the project is held by Nojoom Project Company. The ownership of the project can be proven through:

- Concession agreement signed by the Department of Municipalities and Transport and Nojoom

- Commissioning documents

1.9 Project Start Date

Project start date	06-March-2024
Justification	The project start date is the date on which the project began generating GHG emission reductions.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	06-March-2024 to 05-March-2034

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
06-March-2024 to 31-December-2024	41,999
01-January-2025 to 31-December-2025	119,132
01-January-2026 to 31-December-2026	123,051
01-January-2027 to 31-December-2027	123,051
01-January-2028 to 31-December-2028	123,051
01-January-2029 to 31-December-2029	123,051
01-January-2030 to 31-December-2030	123,051
01-January-2031 to 31-December-2031	123,051
01-January-2032 to 31-December-2032	123,051

01-January-2033 to 31-December-2033	123,051
01-January-2034 to 31-December-2034	81,052
01-January-2035 to 05-March-2035	3,919
Total estimated ERRs during the first or fixed crediting period	1,230,514
Total number of years	10
Average annual ERRs	102,543

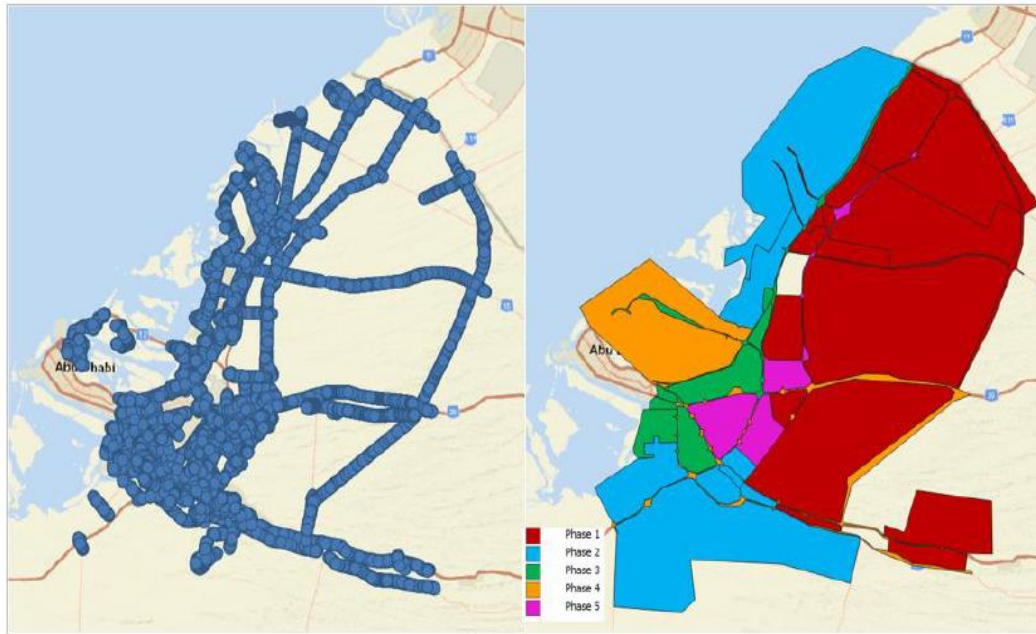
1.12 Description of the Project Activity

The Noor 2 Abu Dhabi Street Lighting Project involves the replacement of 133,473 existing HPS and metal halide MH luminaires with energy-efficient LED luminaires across Abu Dhabi's urban roads and highway areas. This project will significantly reduce GHG emissions by lowering energy consumption through more efficient lighting technology. Additionally, the project will be implemented in five phases, covering small geographical zones over a 24-month period.

The LED luminaires being installed have a wattage range of 38W to 524W and are expected to have a lifespan of over 50,000 operational hours, according to manufacturer specifications. The equipment used for the project includes Schreder models Avento, Izylum, and Ampere Maxi, with a power factor greater than 0.9 and a rated lumen output exceeding 125 lm/w. The LED luminaires will be connected to a Project Lighting Management System (PLMS) for real-time monitoring and control, allowing for optimization of energy use and timely maintenance.

Prior to this project, the existing lighting network utilized HPS/MH luminaires, which were less energy-efficient and had shorter lifespans. In the baseline scenario, these luminaires consumed significantly more electricity. The LED installations will improve energy efficiency, lower maintenance needs, and reduce GHG emissions compared to the baseline scenario.

1.13 Project Location



The project will encompass Abu Dhabi Mainland, covering both urban roads and highways at approximately 24° 27'38"N, 54° 44'48"E. The area has been divided into five distinct phases, each represented by a specific color on the map to the right. These phases are further subdivided into smaller geographical zones to ensure a structured implementation.

1.14 Conditions Prior to Project Initiation

Prior to the initiation of the Noor 2 Abu Dhabi Street Lighting Project, the existing lighting network across the Abu Dhabi Mainland consisted mainly of HPS and MH luminaires. These luminaires were energy-inefficient, consumed higher electricity, and had shorter operational lifespans. It is worth noting that the project was not implemented for the purpose of generating GHG emissions for subsequent reduction. Rather, it aims to replace outdated HPS and MH lighting with more efficient LED technology to achieve real GHG reductions.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with relevant local and national laws, statutes, and regulatory frameworks. The Project complies with all applicable laws such as the Environmental laws of Abu Dhabi. The project has already received the permit for construction and operation by the Department of Municipalities and Transport (DMT). This ensures that the project is aligned with the required legal and regulatory standards for infrastructure projects in Abu Dhabi.

The project has received over 50 electronic No Objection Certificates (e-NOCs) from a wide range of government and non-government entities in the project area. These approvals demonstrate strong support for the project and confirm its compliance with local regulations and requirements.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

The project is not receiving or seeking credit for reductions and removals from another GHG program

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

The project has not been registered and is not active in any other GHG program.

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

There have been no instances where this project was submitted and subsequently rejected by any other GHG programs.

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

The project reductions activities are not included in any emissions trading program or any binding emission limits.

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

This project has no double claiming with other forms of environmental credit and has not sought, received, or planned to receive credit from any other GHG-related environmental credit system.

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☐ No

The project activities do not affect the emissions footprint of any products or services that are part of a supply chain. The emissions reductions are related to the operational energy savings of the lighting systems, not to the production, transport, or supply chain of goods and services.

1.18 Sustainable Development Contributions

The project activity of installing LED luminaires aims to deliver substantial electricity savings, with an estimated reduction of approximately 74% in power consumption over its duration. This significant reduction translates to nearly 2,500 million kWh of electricity saved, highlighting the project's potential for meaningful impact.

By replacing outdated lighting technologies with modern, energy-efficient LED solutions, the project not only enhances energy efficiency but also directly contributes to environmental sustainability. The shift to LED lighting reduces overall energy usage and substantially lowers carbon emissions, aligning with global efforts to combat climate change. Furthermore, this transition supports the long-term goal of creating a more resilient infrastructure while fostering a healthier environment for future generations. Overall, the project represents a crucial step towards achieving Sustainable Development Goals (SDGs) by promoting sustainable practices in urban development as highlighted in the table below:

SDG	SDG Overview	Action Taken
08: Decent Work and Economic Growth	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	Temporary jobs created during the construction phase of the project and permanent jobs created for the operation and maintenance phase
09: Industry, Innovation and Infrastructure	Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation	Upgrading Abu Dhabi's road infrastructure by retrofitting street lighting with advanced energy-efficient LED luminaires. This reduces energy consumption, lowers emissions, and supports sustainable urban development with no compromise on road safety
11: Sustainable Cities and Communities	Make cities and human settlements inclusive, safe, resilient and sustainable	Retrofitting streetlights with energy-efficient LED luminaires reduces emissions through lower energy use, and promotes sustainability by extending the lifespan of lighting infrastructure, contributing to a resilient and eco-friendly urban environment
12: Responsible Consumption and Production	Ensure sustainable consumption and production patterns	Retrofitting streetlights with energy-efficient LED luminaires reduces energy consumption, lowers carbon emissions, and minimizes resource usage, promoting responsible and sustainable infrastructure management
13: Climate Action	Take urgent action to combat climate change and its impacts	Retrofitting streetlights with energy-efficient LED luminaires reduces energy consumption and carbon emissions, helping mitigate climate change and supporting the city's efforts toward a more sustainable, low-carbon future

Contribution to National Priorities

The project aligns with the Abu Dhabi Climate Change Strategy, which targets a 22% reduction in carbon emissions in the Abu Dhabi emirate by 2027. By replacing energy-intensive luminaires with energy-efficient LED technology, the project helps reduce greenhouse gas emissions, supporting both local and global climate goals. It also contributes to the objectives of the Paris

Agreement, which aims to limit global temperature rise to 1.5°C. Furthermore, the project is in line with the UAE's Nationally Determined Contribution (NDC)

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

As per section 6.2 paragraph 51.a. of the applied methodology, in case the project activity involves the replacement of equipment, and the leakage effect of the use of the replaced equipment in another activity is neglected, because the replaced equipment is scrapped, an independent monitoring of scrapping of replaced equipment was implemented.

Since this project involves a one-to-one replacement of street luminaires, with the old luminaires being discarded, no leakage emissions are generated. Therefore, a leakage management plan is not necessary.

1.19.2 Commercially Sensitive Information

The input parameters considered for the investment analysis are deemed commercially sensitive and are classified as confidential information specific to the project proponent. These parameters play a critical role in shaping the investment evaluation process and are integral to the proponent's strategic decision-making framework. These details are not disclosed publicly to protect the project proponent's competitive position and intellectual property.

1.19.3 Further Information

N/A

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The stakeholders were categorized into several key groups:	
	Clients	Department of Municipalities and Transport Abu Dhabi Municipality Abu Dhabi Investment Office
	Shareholders	EDF Engie

	Suppliers and subcontractors	Emirates Electrical & Instrumentation Schreder Revetec Ecyclex Engie Solutions
	Government agencies and national ministries	Environment Agency Abu Dhabi TAQA Department of Energy - Abu Dhabi Ministry of Energy and Infrastructure Ministry of Climate Change and Environment UAE Carbon Alliance
	Broader private sector and wider community	AD Ports Kizad Dolphin Energy Aldar MCC Etisalat Du Bunya Tabreed
Legal or customary tenure/access rights	Not applicable.	
Stakeholder diversity and changes over time	The stakeholder groups consist of individuals from diverse social, economic, and cultural backgrounds, and their make-up is expected to remain the same over time.	
Expected changes in well-being	The project focuses on energy efficiency and delivers various social benefits, including enhanced city lighting and the creation of job opportunities.	
Location of stakeholders	The stakeholders are located in the United Arab Emirates, mainly in the Emirate of Abu Dhabi.	
Location of resources	Not applicable	

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	24-February-2025
Stakeholder engagement process	After identifying the stakeholders, an email invitation was prepared in both Arabic and English. The invitation included an introduction to the project and details of the stakeholder consultation, such as the date, time, and venue. It was sent out two weeks before the event, and an agenda was shared with confirmed attendees one week prior to the event.
Consultation outcome	Stakeholders were pleased with the project's implementation. The project's approval and awarding by Abu Dhabi Municipality and the Department of Municipalities and Transport confirm its compliance with all relevant laws and regulations safeguarding workers' rights in the host country.
Ongoing communication	Ongoing communication with stakeholders was facilitated through multiple channels. Contact emails of key personnel were shared with stakeholders to address any grievances or provide feedback. Additionally, stakeholders were informed about a grievance register that would be accessible on the project proponent's website, offering a formal platform for submitting and tracking concerns.
Stakeholder input	The stakeholders were supportive of the project, appreciating its well-structured planning and the environmentally friendly approach to recycling and safely storing the old bulbs and installations. Most of the questions received focused on the project's registration process under the standard and the associated credits, rather than the project's design itself. As a result, no changes to the project design were deemed necessary.

2.1.3 Free Prior and Informed Consent

Obtaining consent	Electronic No Objection Certificates (NOCs) were requested from a range of stakeholder organizations prior to the commencement of the project. The process involved coordinated communication and engagement with a diverse group of stakeholders, which helped to address any concerns and confirm that the project met necessary regulatory and operational requirements before moving forward with implementation. There are no ongoing or unresolved conflicts regarding the project.
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Outcome of FPIC	More than 50 electronic NOCs were successfully secured from various stakeholder organizations before the project began. These NOCs played a crucial role in confirming that all pertinent parties had thoroughly reviewed and approved the project's scope and potential impacts. The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.
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2.1.4 Grievance Redress Procedure

Development process	Stakeholders were given the contact emails of key personnel to facilitate direct communication and feedback. To support a more structured approach, a grievance register template was developed, capturing important information such as the date of submission, the nature of the grievance, and the steps taken toward resolution, including the resolution date. This template was made available on the project proponent's website, creating a formal platform for submitting and tracking grievances. This process ensures that all concerns are handled efficiently and resolved in a reasonable timeframe, with sensitivity to culturally appropriate conflict resolution methods.
Grievance redress procedure	Ongoing communication with stakeholders was facilitated through multiple channels. Contact emails of key personnel were shared with stakeholders to address any grievances or provide feedback. Additionally, stakeholders were informed about a grievance register that would be accessible on the project proponent's website, offering a formal platform for submitting and tracking concerns.

2.1.5 Public Comments

The public comment period for this project is not yet completed. This section will be updated once it is complete.

Comments received	Actions taken
To be updated	To be updated

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The Noor 2 Abu Dhabi Street Lighting Project is being implemented by Nojoom, a consortium established by EDF and ENGIE. Both companies are global leaders in low-carbon energy generation, with a combined workforce of over 97,000 professionals operating across diverse sectors.

For carbon certification, Nojoom is supported by elementsix, a Dubai-based carbon consultancy with over five years of expertise in project registration and certification. The elementsix team brings extensive experience in registering various emission reduction projects, including renewable energy generation and energy efficiency initiatives. Their knowledge across different projects, regions, and standards ensures the effective management of project registration while adhering to the highest standards of carbon certification.

This collaboration guarantees proper project oversight and compliance with the highest standards of carbon certification.

2.2.2 Risk Assessment

Activities proposed by Nojoom project did not require an environmental impact assessment, nevertheless the following risks were identified, and mitigation or preventative measures are proposed

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	From our analysis, only positive impacts in terms of stakeholders' wellbeing was identified with the project activities
Risks to stakeholder participation	No risk identified	All the stakeholders are identified and proper communication channels are maintained to ensure agreement between Nojoom and the stakeholders thus avoiding any potential risk
Working conditions	Since the project activity involves high voltage operations at an elevation, it is considered as a medium risk activity	Nojoom provided personal protective equipment to the operations team, including helmets, harnesses, gloves, etc. to always ensure safety. All project employees have an employment contract and comply with local regulations

Safety of women and girls	No risk identified	Although Nojoom's project activities are mostly executed by men. The project will ensure gender equality.
Safety of minority and marginalized groups, including children	No risk identified	There are no marginalized groups in the project area and the Nojoom project does not allow child labor.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	Since the project is a replacement of old luminaires, there are no chemical products used. Furthermore, the old luminaires are recycled by an external company to ensure proper disposal.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ²	Mitigation or preventative measure(s) taken
Discrimination	No case of harassment within the operator of the Nojoom project has been reported. The company complies with international standards on this matter and project will ensure that this continues	Not applicable
Sexual harassment	No risk identified	The project complies with all local regulations and rules that prevent and ban sexual harassment.
Equal pay for equal work	No risk identified	The project complies with local ruling on labor laws and employment contracts, in

² The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

		particular the Part-time/Fixed-term employment
Gender equity in labor and work		
Forced labor	No risk identified	The project complies with all UAE laws that ensure no child and forced labor, abolishes human trafficking and complies with the highest global standard of operations. To date, no complaint exist against Nojoom in this regard
Child labor	No risk identified	The project complies with all UAE laws that ensure no child and forced labor, abolishes human trafficking and complies with the highest global standard of operations. To date, no complaint exist against Nojoom in this regard
Human trafficking	No risk identified	The project complies with all UAE laws that ensure no child and forced labor, abolishes human trafficking and complies with the highest global standard of operations. To date, no complaint exist against Nojoom in this regard

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project complies with all national human right laws and follows the highest standard of international human right laws.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
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No risk identified	In the project area there are no sites with indigenous peoples nor areas of cultural heritage.
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2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project activity involves only replacement of old luminaires hence, the land ownership in the project area remains unchanged and is under the ownership of Abu Dhabi municipality.

2.3.5 Benefit Sharing

N/A. Since the project does not involve land sharing/asset sharing as the property belongs to the client i.e. Abu Dhabi Municipality, there are no benefit sharing as such.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	The project activity involves replacement of old luminaires hence, no physical change to the infrastructure is made to impact biodiversity.
Soil degradation and soil erosion	No risk identified	The project activity involves replacement of old luminaires hence, no infrastructural changes are made to impact the soil
Water consumption and stress	No risk identified	The project activity involves replacement of old luminaires hence, no water consumption is involved thus it does not present any water stress.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes

☒ No

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified.	No activity was carried out during the project development that will have an adverse impact on habitats for rare, threatened, or endangered species.
Areas for habitat connectivity	No risk identified.	No activity was carried out during the project development that will have an adverse impact on areas needed for habitat connectivity.

2.4.2 Introduction of Species

N/A, the project only promotes the use and regeneration of native species.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
Not applicable			

2.4.3 Ecosystem Conversion

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	No risk identified	The project will conserve the same ecosystem during its lifetime.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Small-Scale Methodology	AMS-II.C.	Demand-side energy efficiency activities for specific technologies	16.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
AMS-II.C.	This methodology is only applicable if the service level (e.g., rated capacity or output) of the installed, project energy-efficient equipment is between 90% and 150% of the service level of the baseline equipment. Examples of service levels are light output for lighting equipment, water output and temperature for water heating systems, and rated thermal output capacity of air conditioners. The relationship of the service level of the project energy-efficient equipment to the baseline equipment can be one-to-one replacement (e.g., replacement of an inefficient refrigerator with a new and efficient refrigerator) or many-to-one (e.g., replacement of small multiple chillers with a central chiller plant). In the latter case, the service level of the project and baseline can be compared on an aggregate basis.	The project complies with the VCS methodology, as the installed LED luminaires meet the 90%-150% service level requirement compared to the baseline. Strict UAE regulations ensure these luminaires achieve the necessary performance standards for energy efficiency.
AMS-II.C.	Requirements pertaining to the baseline of the retrofit projects and projects involving capacity increase are indicated in paragraphs 20 to 21 in the above-cited	This requirement is not applicable to the project, as it does not involve a capacity increase. The project focuses solely on the replacement of existing luminaires with energy-efficient LED

	general guidelines to SSC CDM methodologies. In the event that project output in year y is greater than the average historical output (average of the three most recent years prior to the project implementation) and the demonstration of the baseline for the incremental capacity is not undertaken, the value of the output in year y is capped at the value of the historical average output level.	luminaires, without increasing the overall capacity of the lighting system. Therefore, no baseline demonstration for incremental capacity is required.
AMS-II.C.	The project lamps shall have a luminous efficacy of no less than 70 Lumens per Watt.	The project lamps achieve a luminous efficacy of 125 lumens per watt, which is well above the 70 lumens per watt minimum requirement.
AMS-II.C.	This methodology credits emission reductions only due to the reduction in electricity and/or fossil fuel consumption from the use of more efficient equipment. However, the calculation of project emissions shall include any incremental emissions, as compared to the baseline, associated with refrigerants used in the project equipment.	This methodology requirement is not applicable to the project as no refrigerants are used in the installed LED luminaires. Therefore, no incremental emissions associated with refrigerants need to be calculated or compared to the baseline.
AMS-II.C.	The aggregate energy savings by a single project may not exceed the equivalent of 60 GWh per year for electrical end-use	The project is divided into 5 phases, and each of these phases does not exceed the energy savings limit of 60 GWh per year for electrical

	energy efficiency technologies. For fossil fuel end-use energy-efficient technologies, the limit is 180 GWh thermal per year in fuel input.	end-use energy efficiency technologies.
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3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Fossil fuel-based power plants serving the grid	CO ₂	Yes	Major source of emissions
		CH ₄	No	Minor source of emissions
		N ₂ O	No	Minor source of emissions
Project	Fossil fuel-based power plants serving the grid	CO ₂	Yes	Major source of emissions
		CH ₄	No	Minor source of emissions
		N ₂ O	No	Minor source of emissions



3.4 Baseline Scenario

In line with the applied methodology AMS-IL.C., version 16.0, the baseline scenario for the Noor 2 Abu Dhabi Street Lighting Project assumes that, without the intervention of the project, lighting would continue to be provided by the existing high-energy-consuming luminaires. This includes the continued use of 133,473 high-pressure sodium (HPS) and metal halide (MH) luminaires to provide street lighting in the project area.

The project activity involves replacing these HPS and MH luminaires with advanced, energy-efficient LED luminaires. The baseline scenario therefore reflects the emissions associated with the operation of the HPS and MH luminaires, which would have continued to operate in the absence of the project.

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☐ No

3.5.2 Additionality Methods

The stepwise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:

Step 0: Demonstration whether the project activity is the first-of-its-kind

According to the CDM – Methodological Tool “Tool for the demonstration and assessment of additionality”, Version 7.0.0”, this step is optional. Therefore, no analysis or evidence will be provided under this step. The project proponent will prove the additionality starting from Step 1 below.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity- Identify realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed CDM project activity.

The project activity reduces electricity consumption by using energy-efficient LED streetlights, which lowers the overall demand on the national electricity grid. Hence, the following alternatives are considered:

Alternative 1: The project activity undertaken without being registered under VERRA.

The option of project activity being undertaken without being registered under VERRA is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as deployment of the project activity faces significant financial barriers established in the subsequent section. The electricity saved from the energy efficiency LED streetlight project would reduce demand on the Abu Dhabi grid, which is supplied by both existing and new fossil fuel-based power plants that would have operated to meet the demand in the absence of the project activity.

Alternative 2: Continuation of the current situation (Without the project activity, the baseline scenario of less energy-efficient LED lights would have required additional electricity from the grid, relying on existing power plants and new capacity expansions to meet the demand)

In this scenario, the project proponent would have opted out of investing in the energy-efficient LED streetlight project, continuing with conventional lighting systems resulting in higher GHG

emissions. Therefore, the use of less energy-efficient LED lights can be considered an appropriate, realistic, and credible baseline scenario for the energy-efficient LED streetlight project.

Outcome of Sub-step 1.a: Identified realistic and credible alternative scenario(s) to the project activity.

Of the two alternatives outlined above, the first alternative is not realistic as project activity as it is not viable without the benefits of carbon credits. The second alternative on the other hand is the baseline scenario for the project activity as per methodology. Thus, continuing the current situation is the most likely alternative to the proposed project activity.

Sub-step 1b: The alternative(s) shall be in compliance with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g., to mitigate local air pollution.

The implementation of the project activity is consistent with mandatory laws and regulations in the UAE and relevant clearances and approvals have been obtained.

Outcome of Sub-step 1.b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country decisions on national and/or sectoral policies and regulations.

Both the alternatives identified above are found to be in compliance with the mandatory laws and regulations taking into account the enforcement of the legislations in the region/country and relevant decisions on national and/or sectoral policies and regulations.

However, Alternative 2 “Continuation of the current situation” has been selected as the appropriate baseline alternative for this project activity in line with the methodology.

Step 2: Investment analysis

An investment analysis is carried out to determine whether the project activity is economically or financially less attractive than at least one other alternative, identified in step 1, without the revenue from the sale of carbon credits. This is demonstrated in line with the tools for the relevant sections as per “Investment Analysis” (Ver 14.0).

Sub-step 2a: Determine appropriate analysis method

A simple cost analysis (Option I) cannot be used as the analysis method as the sale of electricity savings will result in a revenue stream during the project operation. In the absence of the project activity, the continued usage of the less efficient LED lights would have been the most plausible option, which does not require an investment.

Hence an investment comparison analysis (Option II) is also not appropriate for the project activity.

After eliminating Option, I and Option II, the use of a Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be applied in context of the project activity.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using the Benchmark analysis approach (Option III) has been chosen.

Further, this method illustrates that the assessment of the project by the project proponent was carried out prior to the decision to undertake the project was taken and management approval had been granted.

Choice of Financial Indicator:

According to the “Tool for demonstration and assessment of Additionality” Version 07.0.0, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for the analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for the project activity has been chosen as post tax equity IRR.

According to Methodological Tool27: Investment Analysis Version 14.0, the Default Value Benchmark of Equity IRR in the UAE under Group 1, Page 15 (Energy Industries, Energy Distribution, Energy Demand, Waste Handling and Disposal) is 6.55%.

The input parameters for the investment analysis are shown below:

Details of the project	Value
State where the project is situated	Abu Dhabi, UAE
Date of Commissioning - Phase 1	6-Mar-24
Date of Commissioning - Phase 2	6-Jun-24
Date of Commissioning - Phase 3	6-Sep-24
Date of Commissioning - Phase 4	6-Dec-24
Date of Commissioning - Phase 5	6-Mar-25
Concession Period (Yrs.)	12
Phase 1 - Luminaires	28,140
Phase 2 - Luminaires	28,509
Phase 3 - Luminaires	29,292
Phase 4 - Luminaires	19,014
Phase 5 - Luminaires	28,518
Total	133,473

EPC Costs (AED Mn.)	251.48
Other Capex (AED Mn.)	18.04
Total Hard Costs (AED Mn.)	269.53
O&M (AED Mn.)	194.31
Tariff (AED/luminaire)	27.28
Tariff Inflation	2.00%
Cost Inflation	2.00%
Loan Interest Rate	7.00%

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III):

It is calculated based on the cash outflows from and cash inflows into the project activity as shown in the provided spreadsheets.

The IRR and Benchmark analysis shows that the project investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be concluded that the project activity is not the business-as-usual scenario and thus is additional.

Post-Tax Equity IRR	Benchmark IRR
1.69%	6.55%

Sub-step 2d: Sensitivity Analysis

The objective of the sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the scenario presented, in order to provide a cross-check on the robustness of the assumptions used in the development of the investment analysis

The following factor has been subjected to the sensitivity analysis:

1. O&M cost
2. EPC cost
3. Interest Rate

Parameter	Sensitivity Applied	Project Equity IRR	Equity IRR After Negative Adjustment	Equity IRR After Positive Adjustment
EPC Costs (AED Mn.)	±10%	1.69%	7.80%	0.00%
	±5%		5.15%	0.00%
	±1%		2.45%	0.84%
Fixed O&M Costs (AED Mn.)	±10%		6.08%	0.00%
	±5%		4.08%	0.00%
	±1%		2.20%	1.15%
Interest Rate (%)	±2%		7.37%	0.00%
	±1%		4.87%	0.00%

The parameters have been adjusted within a reasonable range to assess their impact in the sensitivity analysis. The findings reveal that the benchmark is only exceeded in scenarios where EPC costs rise by 10% and the interest rate increases by 2%. Both of these situations are highly unlikely within the context of the current project scenario, making such variations improbable.

Step 3: Barrier analysis

Barrier analysis has not been used.

Step 4: Common practice analysis

Common practice analysis has not been used.

3.6 Methodology Deviations

Grid emission factor

The methodology refers to CDM Tool 7 for calculating the electricity grid emission factor. However, as the necessary data is not publicly available, the project owner is currently unable to calculate this factor. Therefore, the project owner requests to use the publicly available grid emission factor for the crediting period as a fixed value.

Based on publicly available information, the following data is available for UAE grid emission factor:

Source	Emission Factor (tCO _{2e} /MWh)	Year
Energy profile, United Arab Emirates published by IRENA	0.467	2022

- Last updated on 31-July-2024 ³		
UNFCCC IFI - United Arab Emirates	0.467	2021

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

According to paragraphs 20 and 21 of the applied methodology AMS-II.C Version 16, Option 1 i.e. ‘Constant Load Equipment’s’ is applicable for the present project. This option applies to equipment that requires the same power (kW) to operate whenever it is energized within specified limits, i.e. are constant load equipment. Baseline energy use and emission are calculated using the following equations:

$$BE_y = (E_{BL,y} \times EF_{CO2,ELEC,y}) + (Q_{ref,BL} \times GWP_{ref,BL})$$

As the project entails replacement of LED in place of ICLs hence no refrigerant is involved. The above equation is then modified as:

$$BE_y = E_{BL,y} \times EF_{CO2,ELEC,y}$$

BE_y = Baseline emissions in year y (tCO₂e)

$E_{BL,y}$ = Energy consumption for the baseline (ICLs) in year y (kWh)

$EF_{CO2,ELEC,y}$ = Electricity emissions factor. If electricity displaced is grid, the emission factor in year y shall be calculated in accordance with the provisions in AMS-I.D (tCO₂/MWh). If electricity displaced is captive electricity, the emission factor in year y shall be calculated in accordance with the “Tool to calculate baseline, project and/or leakage emission from electricity consumption”

Energy consumption for the baseline ($E_{BL,y}$) is calculated using the below formula:

³ https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Middle-East/United-Arab-Emirates_Middle-East_RE_SP.pdf

$$E_{BL,y} = \sum_i \frac{n_i \times \rho_i \times o_i}{1 - l_y}$$

Where:

$\sum_i$ = Sum over the group of i baseline equipment replaced or that would have been replaced. The devices in group i must be closely related by type (in this case luminaires), size, service (e.g. street lighting), and any other relevant factors that determine energy consumption of the equipment

n_i = Number of pieces of equipment of the group of i baseline equipment replaced or that would have been replaced

ρ_i = Electrical power demand (kW) of the group of i baseline equipment

o_i = Average annual operating hours of the group of i baseline equipment

l_y = Average annual technical grid losses (transmission and distribution) during year y for the grid serving the locations where the devices are installed, expressed as a fraction.

4.2 Project Emissions

As per Section 5.5 Project Emission of the applied methodology AMS-II.C Version 16, Project emissions consist of electricity and/or fossil fuel used in the project equipment. Emissions from electricity and/or fossil fuel used in the project equipment is determined as follows:

$$PE_y = (EC_{PJ,y} \times EF_{CO2,y}) + PE_{ref,y}$$

Where:

PE_y = Project emissions in year y (tCO₂e)

$EC_{PJ,y}$ = Energy consumption in project activity in year y. This shall be determined ex post based on monitored values

$EF_{CO2,y}$ = Emission factor for electricity or thermal baseline energy. These are the emissions associated with grid electricity consumption. Given the unavailability of sufficient data to calculate the grid emission factor using TOOL05, the emission factor for the United Arab Emirates used was 0.464 tCO₂/MWh and was sourced from International Renewable Energy Agency (IRENA)

$PE_{ref,y}$ = Since the project involves replacing ICLs with LED lights, no refrigerants are involved. Therefore, there are 0 emissions from refrigerant leakage in this project.

Upon removing the involvement of refrigerants, the equation is modified as below:

$$PE_y = EC_{PJ,y} \times EF_{CO2,y}$$

The project energy consumption in the case of project activities that displace grid electricity is determined as follows using the data of the project equipment or system:

$$EC_{PJ,y} = \sum_i \left(\frac{n_i \times \rho_i \times o_i}{1 - l_y} \right)$$

Where:

n_i = Number of group i project devices operating in time interval t year y

ρ_i = Electrical power demand (kW) of group I project devices measured during the time interval t in year y

o_i = Operating hours of group of I project devices in the time interval t in year y

l_y = Average annual technical grid losses (transmission and distribution) during year y for grid serving the locations where the devices are installed, expressed as a fraction.

4.3 Leakage Emissions

As per section 6.2 paragraph 51.a. of the applied methodology, in case the project activity involves the replacement of equipment, and the leakage effect of the use of the replaced equipment in another activity is neglected, because the replaced equipment is scrapped, an independent monitoring of scrapping of replaced equipment was implemented.

In this project, the baseline lamps were replaced one-to-one with project lamps. During the replacement process, the old luminaires were collected and sent to Ecyclex, an external contractor responsible for recycling the old bulbs. Ecyclex recycled only the bulbs, producing byproducts such as plastics, metals, and fixtures. The remaining components, including the old luminaire shells, cables, and brackets, were returned to the client, Abu Dhabi Municipality, where they were labelled and stored.

As the old luminaires were not transferred to another activity, leakage emissions are not applicable to this project.

$$LE_y = 0$$

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

As per the project activity, replacing old street lighting luminaires with new LED luminaires leads solely to emission reductions, without any removals. The annual emission reductions were calculated using the following formula:

$$ER_y = BE_y - PE_y - LE_y$$

Where:

ER_y = Emissions reduction achieved in year y (tCO₂e)

BE_y = Baseline emissions in year y (tCO₂e)

PE_y = Project emissions in year y (tCO₂e)

LE_y = Leakage emission in year y (tCO₂e)

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
07-March-2024 to 31-December-2024	54,498	12,499	0	41,999	0	41,999
01-January-2025 to 31-December-2025	158,191	39,059	0	119,132	0	119,132
01-January-2026 to 31-December-2026	163,383	40,332	0	123,051	0	123,051
01-January-2027 to 31-December-2027	163,383	40,332	0	123,051	0	123,051
01-January-2028 to 31-December-2028	163,383	40,332	0	123,051	0	123,051
01-January-2029 to 31-December-2029	163,383	40,332	0	123,051	0	123,051
01-January-2030 to 31-December-2030	163,383	40,332	0	123,051	0	123,051
01-January-2031 to 31-December-2031	163,383	40,332	0	123,051	0	123,051
01-January-2032 to 31-	163,383	40,332	0	123,051	0	123,051

December-2032						
01-January-2033 to 31-December-2033	163,383	40,332	0	123,051	0	123,051
01-January-2034 to 31-December-2034	108,885	27,883	0	81,052	0	81,052
01-January-2035 to 05-March-2035	5,192	1,273	0	3,919	0	3,919
Total	1,633,833	403,319	0	1,230,514	0	1,230,514

ii) Complete the table below for the project crediting period. Note that the buffer pool allocation is split proportionally between the estimated reductions and removals. (For example, if a project is estimated to achieve 20,000 tCO_{2e} removals and 80,000 tCO_{2e} reductions and has a buffer contribution of 20%, or 20,000, the estimated removal VCUs would be 16,000 and reduction VCUs would be 64,000).

Vintage period	Estimated baseline emissions (tCO _{2e})	Estimated project emissions (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated buffer pool allocation (tCO _{2e})	Estimated reduction VCUs (tCO _{2e})	Estimated removal VCUs (tCO _{2e})	Estimated total VCU issuance (tCO _{2e})
07-March-2024 to 31-December-2024	54,498	12,499	0	0	41,999	0	41,999
01-January-2025 to 31-December-2025	158,191	39,059	0	0	119,132	0	119,132
01-January-2026 to 31-December-2026	163,383	40,332	0	0	123,051	0	123,051
01-January-2027 to 31-December-2027	163,383	40,332	0	0	123,051	0	123,051

01-January-2028 to 31-December-2028	163,383	40,332	0	0	123,051	0	123,051
01-January-2029 to 31-December-2029	163,383	40,332	0	0	123,051	0	123,051
01-January-2030 to 31-December-2030	163,383	40,332	0	0	123,051	0	123,051
01-January-2031 to 31-December-2031	163,383	40,332	0	0	123,051	0	123,051
01-January-2032 to 31-December-2032	163,383	40,332	0	0	123,051	0	123,051
01-January-2033 to 31-December-2033	163,383	40,332	0	0	123,051	0	123,051
01-January-2034 to 31-December-2034	108,885	27,833	0	0	81,052	0	81,052
01-January-2035 to 05-March-2035	5,192	1,273	0	0	3,919	0	3,919
Total	1,633,833	403,319	0	0	1,230,514	0	1,230,514

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF_{grid}
Data unit	tCO ₂ /MWh
Description	Combined margin emission factor for the grid
Source of data	IRENA UAE Energy profile published 31-July-2024
Value applied	0.464
Justification of choice of data or description of measurement methods and procedures applied	Conservative publicly available information
Purpose of data	Calculation of baseline emissions
Comments	This value will be fixed for the crediting period

Data / Parameter	$\rho_{i,b}$
Data unit	MW
Description	Rated power of baseline lamps
Source of data	Database provided by Abu Dhabi Municipality
Value applied	
Justification of choice of data or description of measurement methods and procedures applied	Existing installed power prior retrofitting is calculated using the latest updated database provided by Abu Dhabi Municipality, after adjustments provided on the Light Technical Clarifications agreed. This value is theoretical and doesn't reflect the real operation of existing luminaires.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	$E_{BL, y}$
Data unit	MWh

Description	Energy consumption for the baseline luminaires in year y
Source of data	Database provided by Abu Dhabi Municipality
Value applied	-
Justification of choice of data or description of measurement methods and procedures applied	Energy consumption prior retrofitting is calculated as the existing installed power prior retrofitting considering the number of lighting hours of the selected year.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	O_i
Data unit	Hours
Description	Operating hours of group of I project devices in the time interval t in year y
Source of data	Astrowatch system
Value applied	4,251
Justification of choice of data or description of measurement methods and procedures applied	The street lighting currently operates based on an astrowatch system, which provides accurate sunrise and sunset timings for each day of the year. By using this data, the total daily operating hours for each luminaire is determined. The astrowatch method ensures consistent and predictable lighting schedules, contributing to the precision of the energy calculations.
Purpose of data	Calculation of project emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$EC_{PJ,y}$
Data unit	MWh
Description	Energy consumption of the project luminaires in year y
Source of data	Measured

Description of measurement methods and procedures to be applied	During the Operations & Maintenance (O&M) of each Phase, the actual energy consumption is captured and presented through the Central Management System (CMS) that gathers the readings from all the nodes in the field. The figures are then matched against one another for comparison and analysis of the Key Performance Indicators (KPI). This system is currently being installed.
Frequency of monitoring/recording	Daily
Value applied	86,922
Monitoring equipment	Each PLMS node, installed at each luminaire, will measure the energy consumption of the luminaire in real time. The CMS will aggregate the data from all PLMS nodes, providing a central platform to monitor and record energy consumption across the entire street lighting network.
QA/QC procedures to be applied	Each PLMS node will be calibrated according to the manufacturer's specifications to ensure accurate energy measurements.
Purpose of data	Calculation of project emissions
Calculation method	Energy consumption is calculated as the installed power after retrofitting considering the number of lighting hours of the selected year.
Comments	

Data / Parameter	O_i
Data unit	Hours
Description	Operating hours of group of I project devices in the time interval t in year y
Source of data	Measured
Description of measurement methods and procedures to be applied	The street lighting currently operates based on an astrowatch system, which provides accurate sunrise and sunset timings for each day of the year. By using this data, the total daily operating hours for each luminaire is determined. During the monitoring phase, it can be monitored through the Project Lighting Management System (PLMS). The Project Company shall provide

	the control nodes as factory-fitted on the LED luminaires via NEMA sockets.
Frequency of monitoring/recording	Daily
Value applied	4,251
Monitoring equipment	Each PLMS node, installed at each luminaire, will measure the operating hours of the luminaire in real time. The CMS will aggregate the data from all PLMS nodes, providing a central platform to monitor and record this data across the entire street lighting network.
QA/QC procedures to be applied	Each PLMS node will be calibrated according to the manufacturer's specifications to ensure accurate energy measurements.
Purpose of data	Calculation of project emissions
Calculation method	
Comments	

5.3 Monitoring Plan

1. Current Energy Consumption Monitoring Methods

While the Central Management System (CMS) and Project Lighting Management System (PLMS) are not yet fully integrated, we are utilizing an alternative method to calculate energy consumption accurately. The following approach is used to measure and record the energy consumption of the luminaires:

- Database of Installed Luminaires:** We have created a detailed database that includes the specific wattage of each luminaire installed in the street lighting network. This database serves as the primary source of information for energy calculations. The wattage values are based on the manufacturer's specifications for each type of luminaire.
- Astrowatch Calculation:** Our street lighting currently operates based on an astrowatch system, which provides accurate sunrise and sunset timings for each day of the year. By using this data, we determine the total daily operating hours for each luminaire. Since we are not using photocells at this stage, the astrowatch method ensures consistent and predictable lighting schedules, contributing to the precision of the energy calculations.

- **Energy Calculation Method:** The energy consumption for each luminaire is calculated using the following formula:

$$\text{Energy Consumption (Wh)} = \text{Wattage of Luminaire (W)} \times \text{Operating Hours (h)}$$

This calculation is performed on a daily basis for each individual luminaire, and the total energy consumption for the entire street lighting network is derived by summing up the energy consumption values for all luminaires.

- **Accuracy:** Since the luminaire database contains precise wattage values and the astrowatch provides accurate operating hours, this method produces energy consumption estimates that are highly reliable until the PLMS and CMS integration is complete.

2. Future Measurement and Recording Methods (Once PLMS and CMS Integration is Complete)

Once the PLMS and CMS are fully integrated, the energy consumption monitoring process will be automated and more granular. The PLMS nodes, which are already installed at each luminaire, will transmit real-time energy consumption data to the CMS. This system will provide direct measurements in watt-hours (Wh), allowing for more precise monitoring, diagnostics, and reporting.

The process will be as follows:

- **PLMS Node Measurements:** Each PLMS node, installed at each luminaire, will measure the energy consumption of the luminaire in real time. These measurements will include voltage, current, power factor, and real energy usage. This data will be transmitted to the CMS at regular intervals.
- **Central Control and Monitoring via CMS:** The CMS will aggregate the data from all PLMS nodes, providing a central platform to monitor and record the energy consumption across the entire street lighting network. The CMS will generate detailed energy reports, offering insights into consumption patterns, potential faults, and efficiency metrics.
- **Energy Reporting:** The CMS will produce energy consumption reports based on the real-time data received from the PLMS nodes. These reports will be available on-demand and can be customized to include different time periods (daily, weekly, monthly) or specific geographic areas.

3. Calibration and Validation Procedures

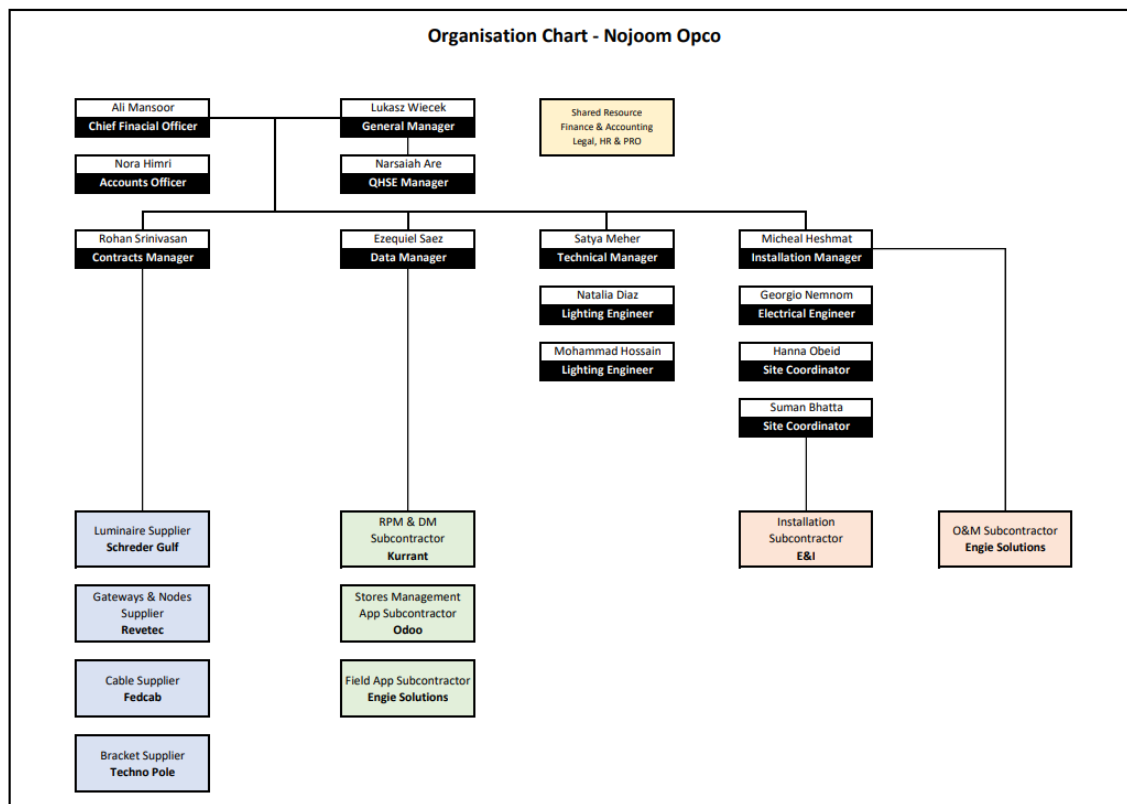
- **PLMS Node Calibration:** Each PLMS node is calibrated according to the manufacturer's specifications to ensure accurate energy measurements. Any nodes showing inconsistent data will be replaced.
- **System-Wide Validation:** The energy consumption calculated by the CMS will be periodically validated by comparing it with sample control cabinet readings and

theoretical system wide energy calculations. This ensures that the system remains accurate over time.

4. Interim Reporting Process

Until the CMS is fully operational, energy consumption reports are generated manually based on the calculated data from the luminaire wattage database and the astrowatch schedules. These reports will continue to be provided at regular intervals to ensure transparency and to allow for ongoing monitoring of energy usage.

The organizational chart for the project is as shown below:



Below is a summary of the responsibilities for each key role in the project:

1. **Lukasz Wiecek (Contractual/Financial Lead):** Manages contract execution and compliance, liaising regularly with ADM, shareholders, and lenders. Key tasks include project coordination, supplier management, ensuring profitability, and reporting.
2. **Ali Mansoor (CFO):** Oversees project finances, managing invoicing, payments, and reporting to shareholders and lenders. Acts as the financial contact for investors, lenders, and authorities, and ensures the project's financial health.

3. **Satya Meher (Technical Manager & Lighting Design Engineer):** Leads the engineering and design office, optimizing lighting design, coordinating with suppliers, and ensuring technical standards. Manages technical KPIs and supports procurement.
4. **Rohan Srinivasan (Contracts & Procurement Manager):** Handles supplier and subcontractor contracts, ensuring compliance with project terms. Manages logistics, sets up KPIs, and oversees warranty terms.
5. **Ezequiel Saez (Data Manager):** Manages data integration with existing systems, ensures data quality and security, and supports reporting and troubleshooting. Oversees data sharing and maintains data standards.
6. **Michael Heshmat (Installation Manager):** Oversees site installation, coordinating resources and schedules, ensuring safety, quality, and regulatory compliance, and managing subcontractors.
7. **Georgio Nemnom, Hanna Obeid, and Suman Bhatta (Work Coordinators):** Handle on-site logistics and staff coordination, monitor progress, ensure compliance, and report to the Installation Manager.
8. **Narsaiah Are (QHSE Manager):** Implements safety, quality, and environmental policies, ensuring compliance with municipal and ISO standards. Provides training, manages audits, and addresses non-conformity events.
9. **Operations & Maintenance Manager:** Leads maintenance operations for lighting systems, monitors performance, organizes permits, and ensures safe and compliant maintenance work on site.

Each role is structured to support seamless project execution, adherence to regulations, and consistent reporting and collaboration with stakeholders.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification
<u>3.5.2</u>	The input parameters considered for the investment analysis are deemed commercially sensitive and are classified as confidential information specific to the project proponent.	These parameters play a critical role in shaping the investment evaluation process and are integral to the proponent's strategic decision-making framework. These details are not disclosed publicly to protect the project proponent's competitive position and intellectual property.